

Stroke Risk Prediction Analysis

Capstone Three: Final Project Report

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I. Problem Statement

According to the World Health Organization, stroke is the third-leading cause of death and disability globally, making it a major public health concern. This project aims to develop a machine-learning model that uses patient demographics, medical conditions, and lifestyle factors—including age, gender, hypertension, heart disease, average glucose level, BMI, and smoking status—to identify individuals who may be at an elevated risk of stroke.

II. Data Description

The stroke prediction dataset was obtained from Kaggle and contains 5,110 patient records. The dataset includes demographic, medical, and lifestyle variables such as age, gender, hypertension, heart disease, average glucose level, BMI, smoking status, work type, residence type, and marital status. The target variable, stroke, indicates whether a patient had experienced a stroke.

Using this dataset, data wrangling, exploratory data analysis, preprocessing, and training-data development were performed. Four classification models were then evaluated: a dummy classifier, logistic regression, a decision tree, and a random forest.

One important limitation of the dataset is its substantial class imbalance. Of the 5,110 records, only 249 represent patients who experienced a stroke, or approximately 4.9% of the dataset. Because stroke cases are relatively rare, accuracy alone could provide a misleading assessment of model performance. Therefore, model evaluation focused on recall, precision, F1-score, PR-AUC, ROC-AUC, and the confusion matrix.

The key beneficiaries of this project include patients, healthcare providers, hospitals, public health organizations, and insurance providers. A stroke-risk prediction model could potentially help identify higher-risk individuals, support preventive

interventions, improve patient outcomes, and reduce the financial burden associated with stroke treatment and long-term disability.

III. Exploratory Data Analysis

Exploratory data analysis was performed to better understand the dataset, identify data-quality issues, examine potential relationships between the predictors and stroke status, and determine appropriate preprocessing and modeling strategies.

The dataset contains 5,110 patient records, of which only 249 (approximately 4.9%) are stroke cases. (Figure 1) This substantial class imbalance is an important consideration for model development because a model could achieve high accuracy by primarily predicting the majority non-stroke class. Therefore, evaluation metrics such as recall, precision, F1-score, PR-AUC, ROC-AUC, and the confusion matrix were used in addition to accuracy.

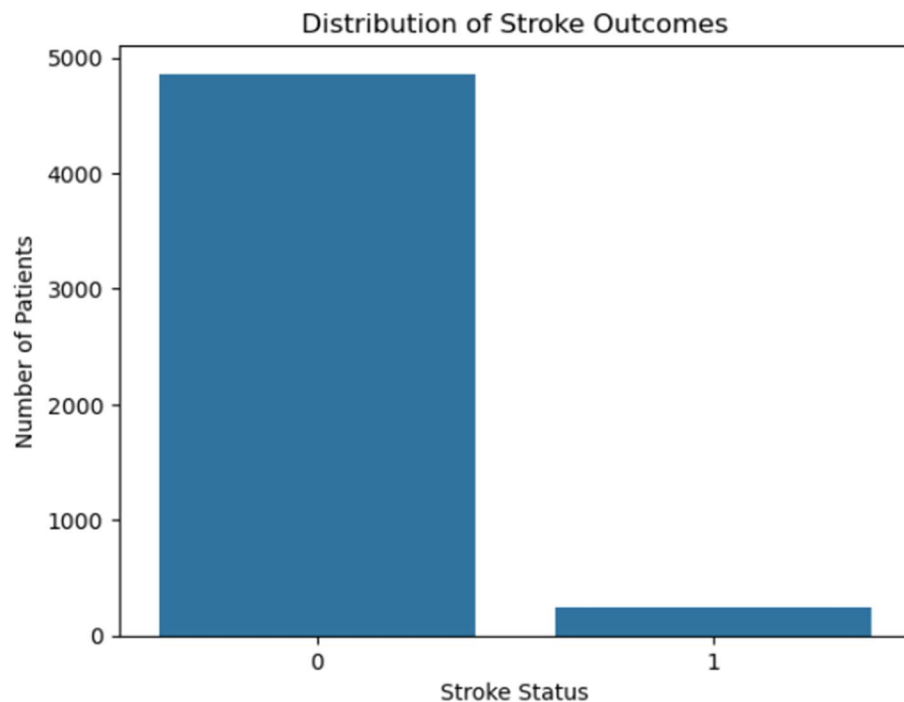


Figure 1. Distribution of Stroke Outcomes (0 = No Stroke, 1 = Stroke)

BMI was the only variable with a notable amount of missing data, with 201 missing values. The analysis also showed that missing BMI values were more common among stroke cases than among non-stroke cases. BMI was missing for about 3.3% of non-stroke records and 16.1% of stroke records. Because BMI may contain useful

predictive information, the missing values were addressed during preprocessing rather than removing the affected records.

Among the numerical features, age showed a clear association with stroke status, with stroke cases occurring more frequently among older individuals. (Figure 2) Average glucose levels also tended to be higher among patients who had experienced a stroke. (Figure 3) BMI showed less separation between the stroke and non-stroke groups.

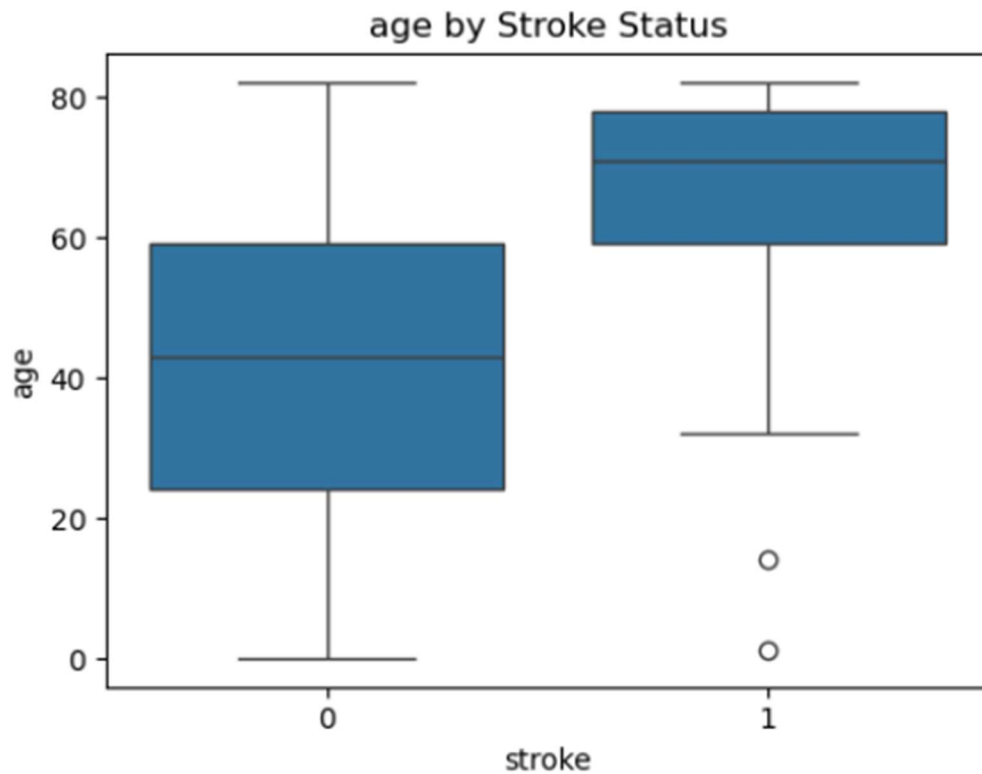


Figure 2. Age Distribution by Stroke Status (0 = No Stroke, 1 = Stroke)

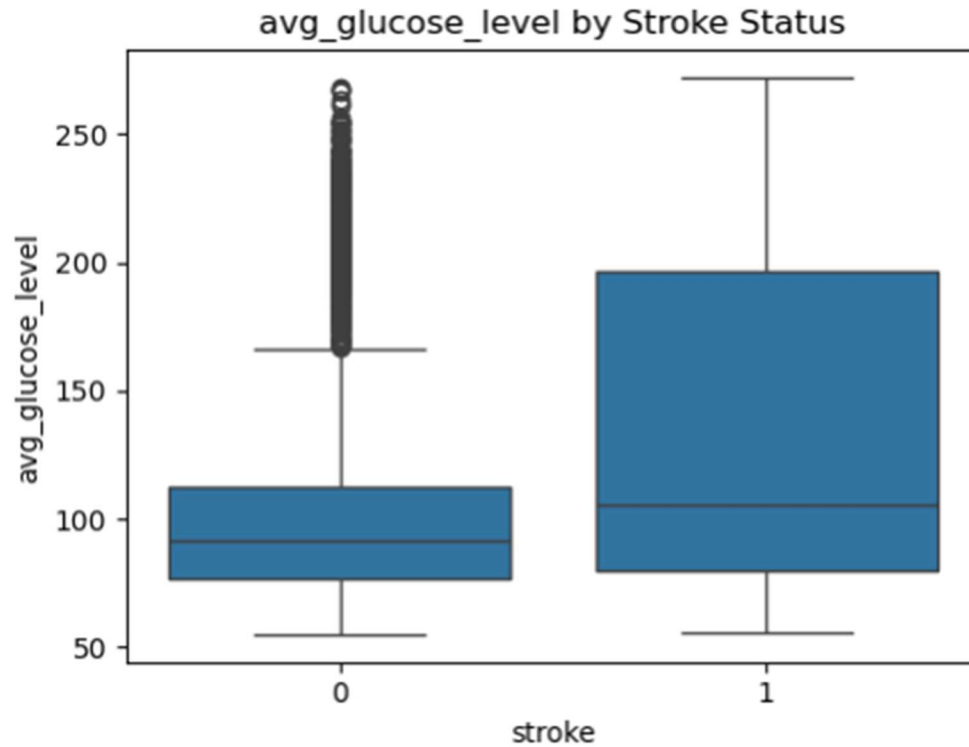


Figure 3. Glucose vs. Stroke (0 = No Stroke, 1 = Stroke)

The categorical analysis showed that hypertension and heart disease were more common among stroke cases. (Figure 4, Figure 5) Relationships were also observed for several other categorical variables; however, some of these patterns may be influenced by age and other related characteristics rather than representing independent relationships with stroke.

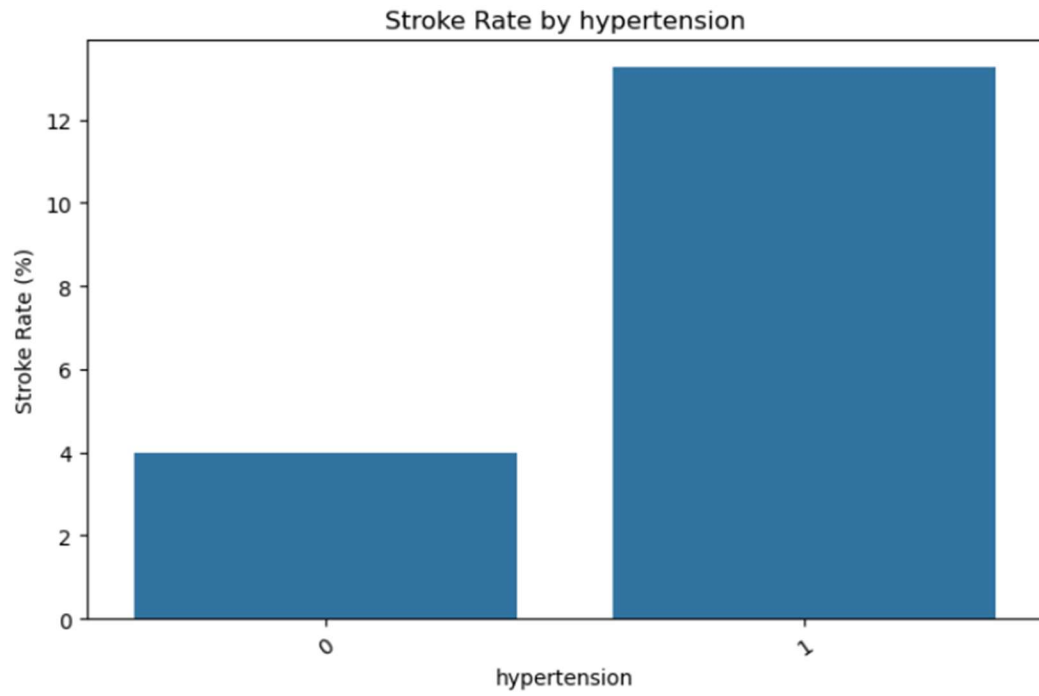


Figure 4. Hypertension vs. Stroke (0 = No Hypertension, 1 = Hypertension)

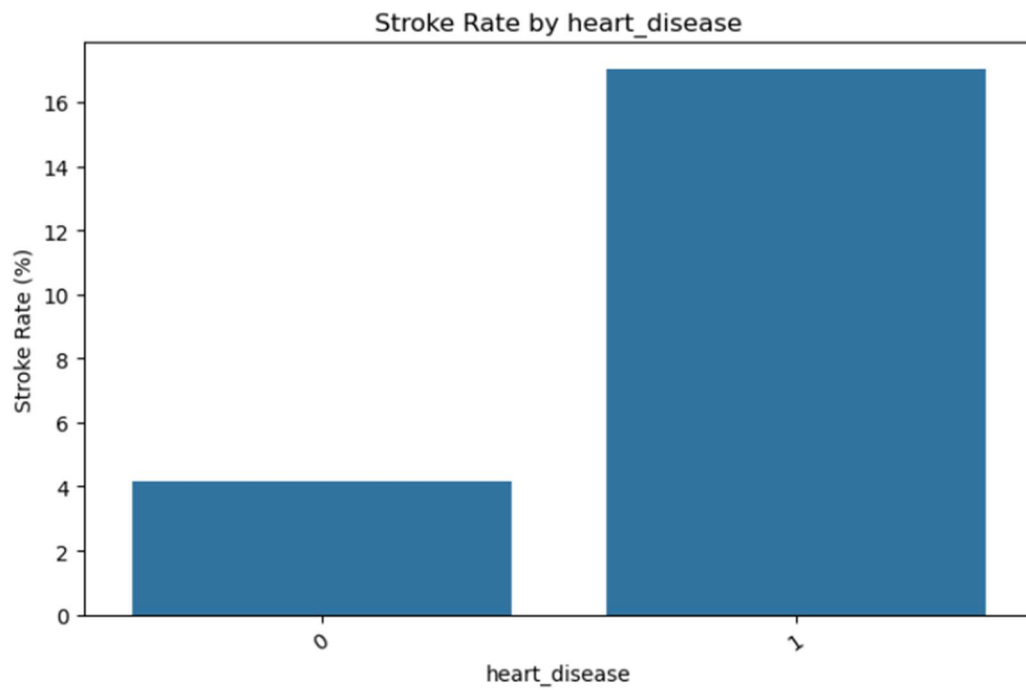


Figure 5. Heart Disease vs Stroke (0 = No Heart Disease, 1 = Heart Disease)

Finally, the id variable was identified as a patient identifier rather than a meaningful predictive feature and was therefore removed before modeling. Overall, the EDA revealed substantial target-class imbalance, missing BMI values, and several potentially informative predictors, particularly age, hypertension, heart disease, and average glucose level.

IV. Preprocessing & Modeling

A. Data Preprocessing

Before model development, the dataset was preprocessed to prepare the features for machine-learning classification while minimizing the risk of data leakage. The id column was removed because it was a patient identifier and did not provide meaningful predictive information. The Unnamed: 0 column, which represented an unnecessary index, and the bmi_missing indicator created during exploratory analysis were also removed. The stroke variable was separated from the remaining features and used as the target variable.

The dataset was then divided into training and test sets using an 80/20 split with a random_state of 42. Because only approximately 4.9% of the records represented stroke cases, stratification was applied using the target variable. This preserved approximately the same proportion of stroke and non-stroke cases in both the training and test sets.

The predictor variables were divided into three groups based on their data types. The numerical features consisted of age, avg_glucose_level, and bmi. Missing numerical values were imputed using the median, after which the numerical features were standardized using StandardScaler.

The categorical features consisted of gender, ever_married, work_type, Residence_type, and smoking_status. Missing categorical values were imputed using the most frequent category, and the variables were transformed using one-hot encoding. The encoder was configured to ignore previously unseen categories during transformation. The binary variables, hypertension and heart_disease, were already represented numerically and were therefore passed through without additional transformation.

These preprocessing steps were combined using a ColumnTransformer and incorporated into the machine-learning pipelines. This approach ensured that preprocessing was applied consistently across models and that transformations

such as imputation and scaling were learned from the training data rather than the complete dataset, helping to prevent data leakage.

B. Model Evaluation and Selection

Four classification models were evaluated: a dummy classifier, logistic regression, a decision tree, and a random forest. Because the dataset was highly imbalanced, model performance was assessed using multiple evaluation metrics rather than accuracy alone. The primary metrics included precision, recall, F1-score, ROC-AUC, and PR-AUC. Five-fold stratified cross-validation was used on the training data so that each fold maintained a similar proportion of stroke and non-stroke cases.

Initial Model Cross-validation Performance

Model	Accuracy	Precision	Recall	F1	ROC_AUC	PR_AUC
Logistic Regression	0.739	0.133	0.789	0.227	0.839	0.193
Random Forest	0.951	0.100	0.005	0.010	0.811	0.163
Decision Tree	0.919	0.151	0.141	0.145	0.550	0.064
Dummy	0.951	0.000	0.000	0.000	0.500	0.049

The dummy classifier served as a baseline model by always predicting the majority class. Although it achieved approximately 95.1% accuracy, its precision, recall, and F1-score for stroke cases were all 0. This demonstrated why accuracy alone was not an appropriate measure for this dataset.

Among the initial models, logistic regression provided the strongest overall performance. It achieved an average cross-validation recall of approximately 0.789, ROC-AUC of 0.839, and PR-AUC of 0.193. Although its accuracy was lower than that of the dummy classifier and random forest, its substantially higher recall showed that it was much more effective at identifying stroke cases.

The decision tree achieved an accuracy of approximately 0.919 but had relatively low recall of 0.141 and a PR-AUC of 0.064. The random forest achieved an accuracy of approximately 0.951 and a ROC-AUC of 0.811, but its recall was only

0.005, indicating that it identified very few stroke cases using the default classification threshold.

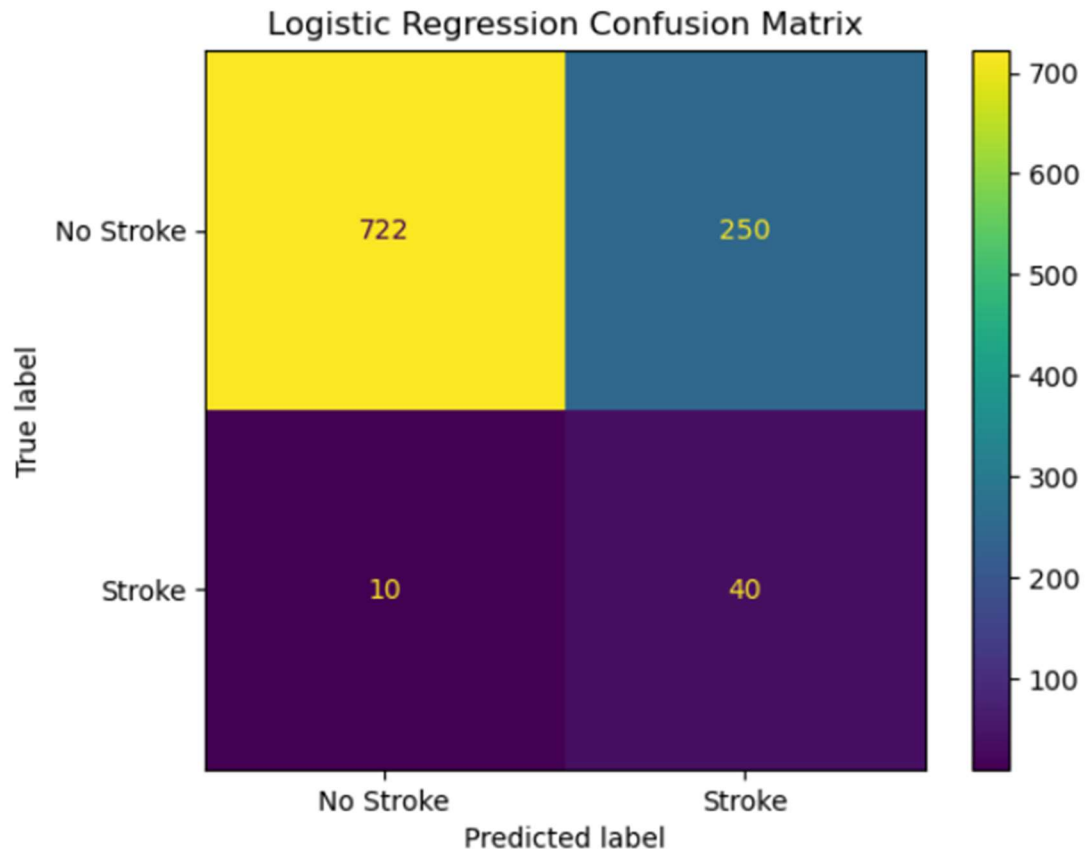
Because logistic regression produced the highest PR-AUC and the strongest recall among the initial models, it was selected as the leading candidate.

Hyperparameter tuning was then performed using GridSearchCV, with PR-AUC (average_precision) as the optimization metric. Logistic regression was tuned across several values of the regularization parameter C. The tuned logistic regression produced essentially the same cross-validation performance as the original model, indicating that the initial parameterization was already near the best-performing configuration within the tested range.

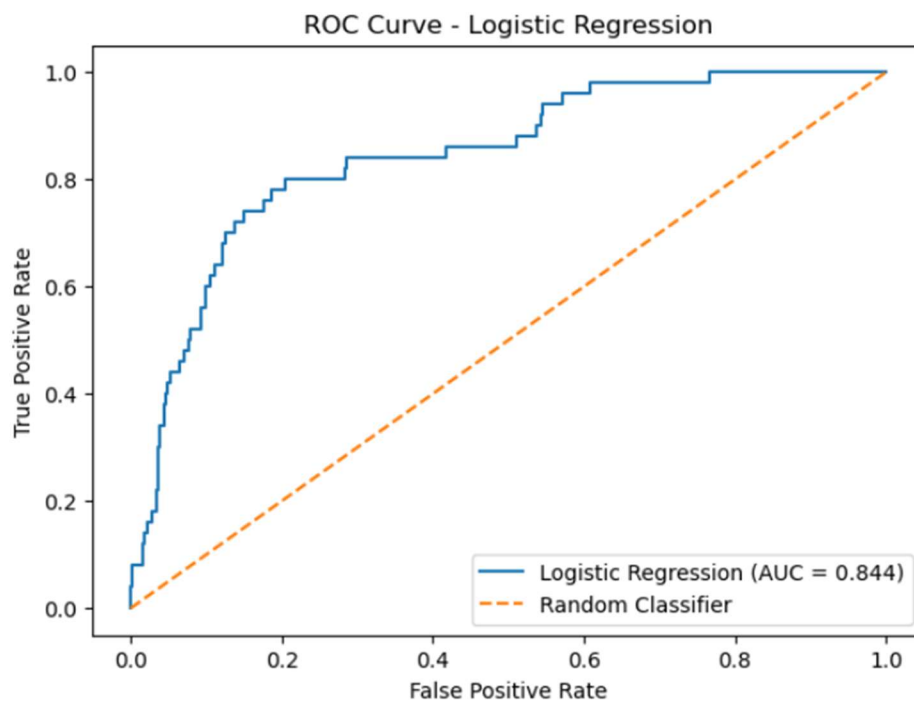
The decision tree and random forest were also tuned using grid search. Tuning substantially improved the recall of both models. The tuned decision tree achieved a recall of approximately 0.628 and a PR-AUC of 0.149, while the tuned random forest achieved a recall of approximately 0.769 and a PR-AUC of 0.180. Despite these improvements, logistic regression continued to provide the highest PR-AUC and slightly higher recall.

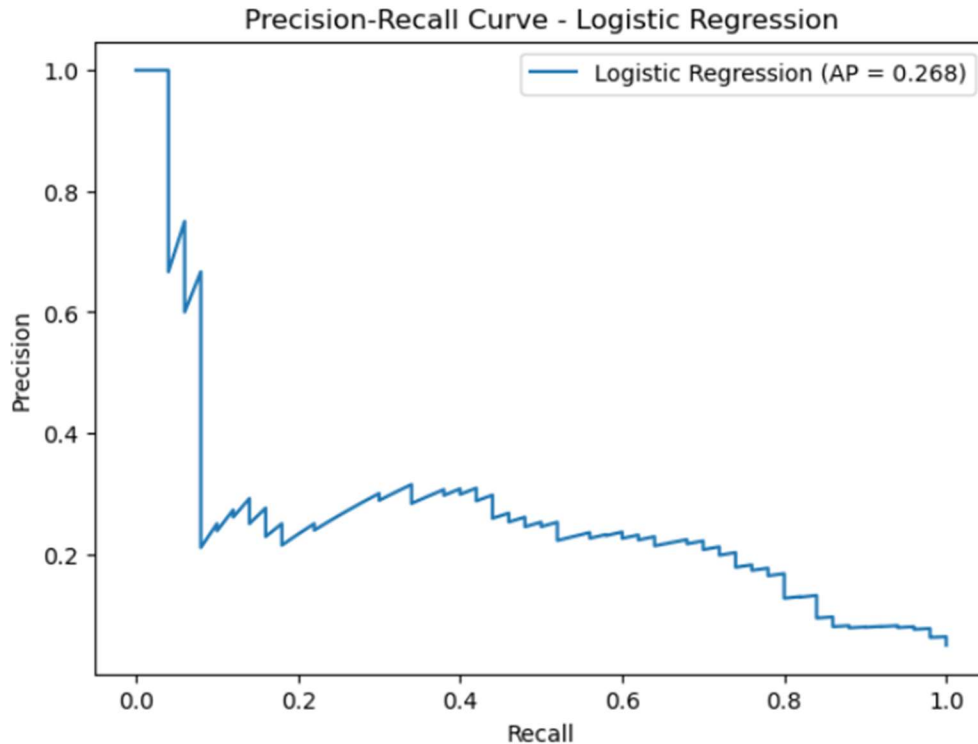
Based on the cross-validation results, logistic regression was selected as the final model because it offered the best overall balance for identifying stroke cases in this imbalanced dataset. In particular, it produced the highest PR-AUC and recall while also maintaining strong ROC-AUC performance.

The selected logistic regression model was then evaluated on the previously unseen test set. It achieved an accuracy of approximately 0.746, precision of 0.138, recall of 0.800, F1-score of 0.235, ROC-AUC of 0.844, and PR-AUC of 0.268. The recall of 0.800 indicates that the model correctly identified approximately 80% of the stroke cases in the test set. However, the relatively low precision indicates that many patients predicted as stroke cases were actually non-stroke cases. This tradeoff may be acceptable in a stroke-risk screening context, where failing to identify a potentially high-risk patient may be more costly than flagging additional patients for further evaluation.



The model correctly identified 40 of the 50 stroke cases in the test set, while missing 10 stroke cases and incorrectly flagging 250 non-stroke cases.





V. Conclusion and Recommendations

A. Conclusion

This project developed and evaluated machine-learning models for identifying individuals who may be at elevated risk of stroke using demographic, medical, and lifestyle characteristics. Exploratory data analysis showed that stroke cases represented only 4.9% of the dataset, creating a substantial class-imbalance challenge. Age showed a clear association with stroke status, while hypertension, heart disease, and higher average glucose levels were also more common among stroke cases.

Four classification models were evaluated: a dummy classifier, logistic regression, a decision tree, and a random forest. Although the dummy classifier and random forest achieved high accuracy, their low recall demonstrated that accuracy alone was not an appropriate measure of performance for this imbalanced dataset. Logistic regression provided the strongest overall balance of recall, PR-AUC, and ROC-AUC and was therefore selected as the final model.

On the unseen test set, the logistic regression model achieved a recall of 0.800, meaning that it correctly identified 80% of the stroke cases. The model also achieved a ROC-AUC of 0.844 and a PR-AUC of 0.268. However, its precision was

relatively low at 0.138, indicating that many individuals classified as potential stroke cases were false positives.

Overall, the results demonstrate that machine-learning methods can identify patterns associated with stroke status even in a highly imbalanced dataset. The selected logistic regression model prioritized identifying stroke cases rather than maximizing overall accuracy. While the model should not be considered a clinical diagnostic tool, it demonstrates the potential value of predictive modeling as a screening approach for identifying individuals who may benefit from further medical evaluation.

B. Recommendations

Based on the results of this project, logistic regression may serve as a useful starting point for stroke-risk screening because of its relatively high recall and its ability to identify a large proportion of stroke cases. In a screening setting, prioritizing recall may be appropriate because failing to identify an individual who is potentially at high risk of stroke could have more serious consequences than referring a lower-risk individual for additional evaluation.

However, the model's relatively low precision indicates that many individuals identified as potential stroke cases would be false positives. Therefore, predictions from the model should not be used independently for clinical decision-making. Instead, a predictive model of this type could potentially be used as an initial screening tool alongside clinical evaluation and other medical information.

Future development should focus on improving precision while maintaining high recall. This could include testing additional machine-learning algorithms, adjusting the classification threshold, exploring alternative approaches for handling class imbalance, and incorporating additional clinically relevant predictors. Model performance should also be evaluated using larger and more representative patient populations before considering real-world implementation.

Overall, the model should be viewed as a decision-support or screening approach rather than a diagnostic tool. With additional data, validation, and model development, stroke-risk prediction models may help healthcare providers identify higher-risk individuals who could benefit from further evaluation and preventive care.